

**SIX WEEKS INDUSTRIAL TRAINING
PROJECT REPORT
ON**

AI_SALARY_PREDICTION_SYSTEM

A Report Submitted in partial fulfillment for the degree of

Bachelor of Technology

in

Computer Science & Engineering



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1.INTRODUCTION

Artificial Intelligence (AI) and Machine Learning (ML) have become essential technologies in today's digital world. They are widely used in industries to automate decision-making and improve efficiency. One common application of Machine Learning is salary prediction, where the salary of an employee is estimated based on different factors such as years of experience.

This project, AI Based Salary Prediction System Using Linear Regression, is developed to predict the expected salary of an employee by analyzing the relationship between years of experience and salary. The project uses the Linear Regression algorithm, one of the most popular supervised machine learning techniques. The complete project includes data analysis, model training, prediction, and a user-friendly Streamlit web application.

2. PROBLEM STATEMENT

Many companies and organizations need to estimate employee salaries during recruitment and performance evaluation. Manual estimation may be inconsistent and can vary from one person to another. A Machine Learning-based salary prediction system provides quick, reliable, and data-driven predictions using historical data.

3. OBJECTIVES

The main objectives of this project are:

To perform Exploratory Data Analysis (EDA) on the salary dataset.

To understand the relationship between years of experience and salary.

To build a Machine Learning model using Linear Regression.

To predict salary accurately based on user input.

To develop a Streamlit-based interactive web application.

To understand the complete Machine Learning project workflow.

4. Project Description

The project uses a dataset containing employee salary information along with years of experience. Initially, Exploratory Data Analysis (EDA) is performed to understand the dataset, identify missing values, detect duplicate records, and visualize the relationship between the variables.

After analyzing the data, the dataset is divided into training and testing datasets. A Linear Regression model is trained using the training data. Once the model is trained successfully, it is saved as a pickle file (salary_model.pkl).

Finally, a Streamlit-based web application is developed where users can enter their years of experience using a slider. The trained model predicts the estimated salary instantly. The application also displays the dataset preview and regression graph for better understanding.

5. Tools & Technologies Used

The following technologies are used in this project:

Python

Pandas

NumPy

Matplotlib

Scikit-learn

Streamlit

Jupyter Notebook

6. Methodology

The project is completed using the following steps:

Collect the salary dataset.

Load the dataset into Jupyter Notebook.

Perform Exploratory Data Analysis (EDA).

Check missing and duplicate values.

Visualize the data using charts.

Split the dataset into training and testing sets.

Train the Linear Regression model.

Evaluate the model performance.

Save the trained model.

Develop the Streamlit web application.

Test the application using different input values.

7. Expected Outcome

The expected outcome of this project is an interactive web application capable of predicting employee salaries based on years of experience. The prediction process becomes faster, easier, and more consistent compared to manual estimation. The project also provides practical knowledge of Machine Learning model development and deployment.

8. Applications

This project can be used in various areas such as:

Human Resource (HR) Departments

Recruitment Agencies

Salary Analysis

Employee Performance Evaluation

Educational Machine Learning Projects

Career Guidance Systems

9. Future Scope

The current project predicts salary using only years of experience. In the future, additional features such as education level, job role, technical skills, company type, industry, certifications, and location can be added to improve prediction accuracy. Advanced Machine Learning algorithms such as Random Forest, Decision Tree, and XGBoost can also be used. The application can be deployed online using cloud platforms so that users can access it from anywhere.

10. Conclusion

The AI Based Salary Prediction System Using Linear Regression successfully demonstrates the complete Machine Learning workflow, including data collection, Exploratory Data Analysis (EDA), model training, evaluation, and deployment using Streamlit. The project provides accurate salary predictions based on years of experience and serves as an excellent learning example for beginners in Artificial Intelligence and Machine Learning. It also highlights the practical use of supervised learning algorithms in solving real-world business problems.